

PART D  
CAP182  
NAME : KHANYISILE LOLWANE  
STUDENT NUMBER : 26304954

According to the models that we have built today, we can conclude that the best model to be considered and applied for the CTR is the second model that we used which is the Hist Gradient Boosting Classifier. It is better suited for the CTR prediction task due to the fact that, it performed better than Model 1 across all evaluation metrics which are Accuracy, AUC-ROC, Precision, Recall, and F1 Score. Model 2 also had a higher Recall of 0.7589 compared to 0.7229 for the first Model, meaning it identified more users who actually clicked on an ad. Its Precision was also higher with 0.2801 compared with 0.2155, meaning its click predictions were more accurate. Model 2 also had a higher F1 Score of 0.4092 compared with 0.3320, showing a better balance between Precision and Recall. In addition, Model 2 had a higher AUC-ROC of 0.7596 compared with 0.6353, meaning it was better at distinguishing between users who clicked and those who did not. The Gradient Boosting Classifier also consisted of a fewer False Positive and False Negatives, therefore we would say that it performed better on our test data compare to Logistic Regression.

The Hist Gradient Boosting Classifier model can be improved by including hyper parameter tuning by testing different settings, such as the number of trees, learning rate and maximum tree depth. This would assist us in finding the best settings to improve the model's predictions. The model can also be improved by adding more useful features, such as device type, time of day, day of the week, previous clicks, previous purchases, product category, search history, and the number of times a product was viewed. These features can help the model better understand user behavior and also predict which products are user friendly and most likely to be clicked on.

Another way to improve the model is by using ensemble methods, where different models, such as Logistic Regression, Gradient Boosting, and Random Forest, are combined. Different models can identify different patterns, so combining them may improve the overall prediction. Lastly, using more data could help the model learn more patterns, historical data about clicks, product views, purchases, searches, and ad interactions could improve its performance.

It is important to note that before using the data in Model 2, the data quality problems must also be handled. Since STADIOalot has inconsistent product categories across millions of products, standardization rules can be used to make category names consistent. Missing values can be handled using appropriate methods, such as filling them with common values or marking them as unknown. Validation checks can also be used to identify incorrect or unreliable information. Outliers should also be detected and checked, especially for values such as prices. These steps will help ensure that the data given to the model is clean and reliable.

And lastly with the Hist Gradient Boosting Classifier Model performing better than the Logistic Regression across all evaluation metrics. Model 2 achieved an AUC-ROC of 0.7596, compared with 0.6353 for Model 1, and also achieved higher Precision, Recall, F1 Score, and Accuracy. Therefore, our findings are consistent with the literature, which shows that models that can capture more complex relationships in advertising data can provide stronger CTR prediction results than simpler models.